Lecture-6
Fluid Kinematics I: Lagrangian and Eulerian Description

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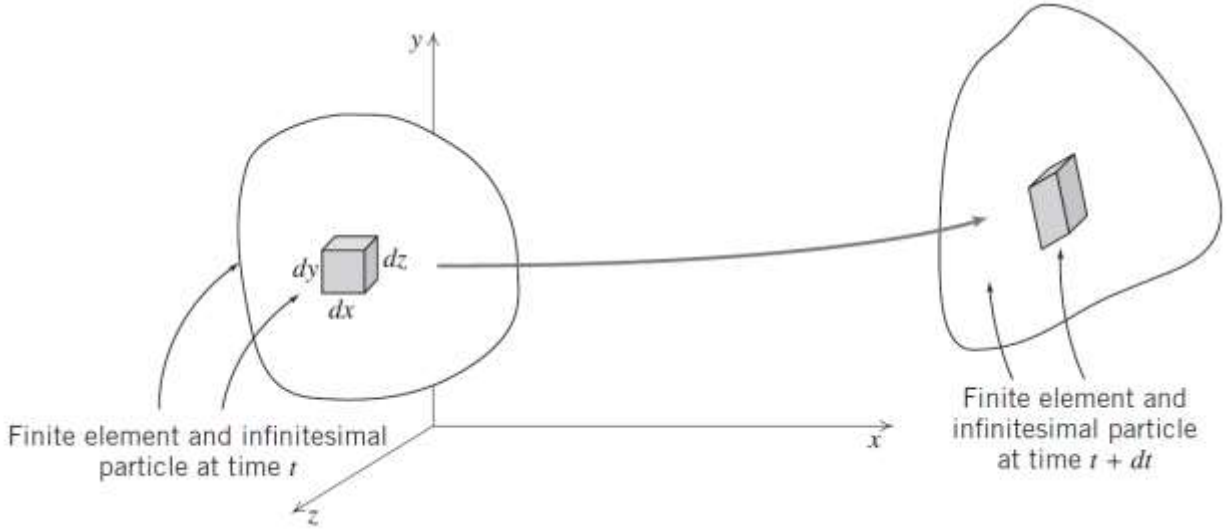
Learning Objectives

By the end of this lecture, you should be able to

1. Understand the **need to study kinematics** for fluids
2. Compare and contrast the **Lagrangian and Eulerian perspectives** for flow analysis
3. Explain the physical and mathematical significance of the **material derivative**
4. Deconstruct the **Eulerian acceleration** of a fluid element into its **local and advective components**

Why should we study kinematics?

- The behavior of a fluid is characterized by its **rate of deformation** in response to an **applied shear stress.**
- Kinematics provides the **mathematical framework to quantify the rate of deformation** of a fluid element**.**
- There are primarily two perspectives to study kinematics, which are called the **Lagrangian perspective** and the **Eulerian perspective.**



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Lagrangian vs Eulerian descriptions

- The Lagrangian description focuses **attention on material particles** as they move through the flow.
- The Eulerian viewpoint has us **watch a fixed point in space** as time proceeds.
- Lagrangian analysis of fluid motion is usually quite difficult and is seldom attempted.

What is the material derivative?

- In the Eulerian viewpoint, we are focused on specific points in space at various times, because of which we **lose the ability to track the history of the particle**.
- The **material derivative is an expression** that allows us to formulate, in Eulerian variables, a **time derivative evaluated as we follow a material particle**.

$$\frac{D\vec{V}}{Dt} \equiv \vec{a}_p = u\frac{\partial \vec{V}}{\partial x} + v\frac{\partial \vec{V}}{\partial y} + w\frac{\partial \vec{V}}{\partial z} + \frac{\partial \vec{V}}{\partial t}$$

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Revisiting the learning objectives

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